

Mathematics for Information Technology

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Room.518 or Room.506 (MIV Lab)

DIFFERENTIATION (I)

Differentiation

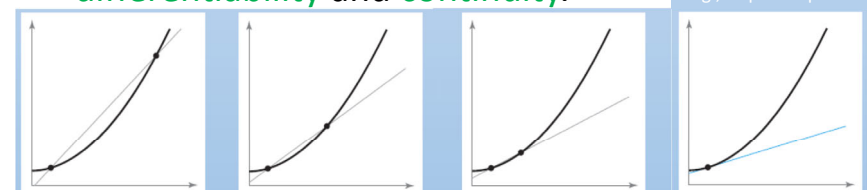
- The most important processes of calculus---*differentiation*.
 - Learn new methods and rules for finding *derivatives of functions*.
 - Apply these rules to find such things as *velocity*, *acceleration*, and the *rates of change* of two or more related variables.

Differentiation

• OBJECTIVES

Learn how to ...

- find the *derivative* of a function using the *limit definition* and understand the *relationship* between *differentiability* and *continuity*.



To approximate the slope of a tangent line to a graph at a given point, find the slope of the secant line through the given point and a second point on the graph.
As the second point approaches the given point, the approximation tends to become more accurate.

Differentiation

• OBJECTIVES

Learn how to ...

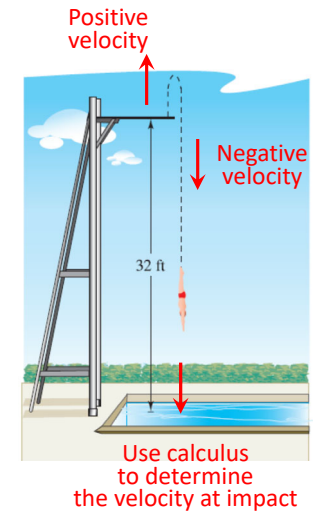
- find the **derivative** of a function using **basic differentiation rules**.
- find the **derivative** of a function using the **Product Rule** and the **Quotient Rule**.

Differentiation

• OBJECTIVES

Learn how to ...

- find a **related rate**.



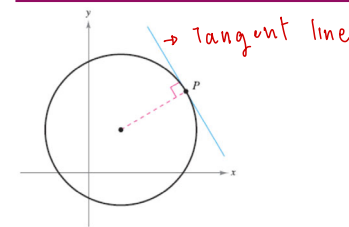
The Derivative and the Tangent Line Problem

I

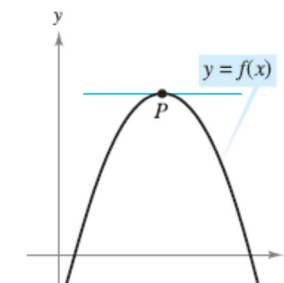
Objectives

- Find the **slope** of the **tangent line** to a curve at a point.
- Use the **limit** definition to **find the derivative** of a function.
- Understand the **relationship** between **differentiability** and **continuity**.

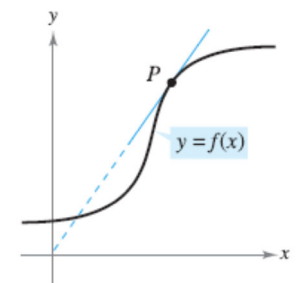
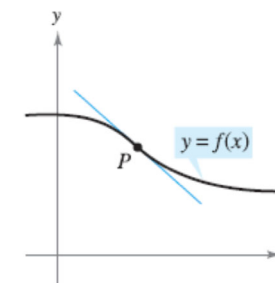
The Tangent Line Problem



Tangent line to a circle

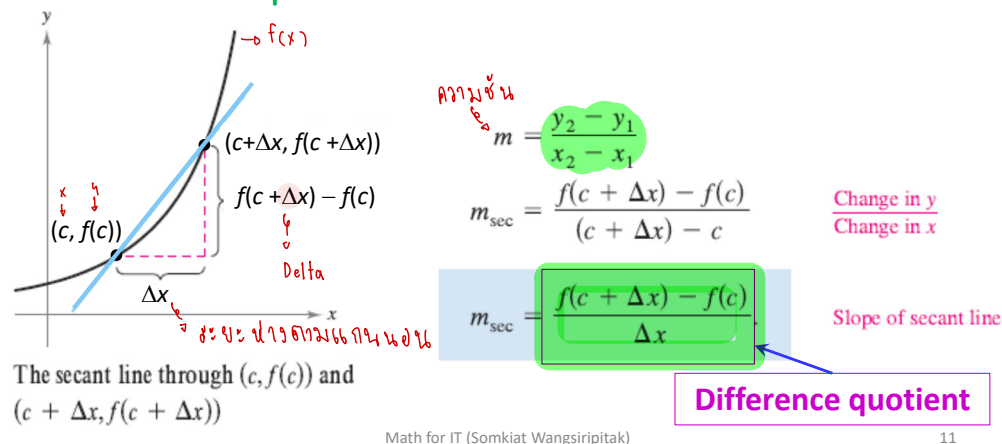


Tangent line to a curve at a point



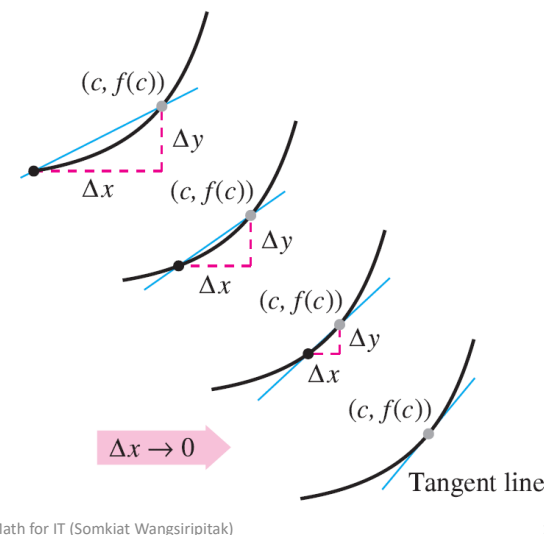
The Tangent Line Problem

- Approximate the slope using a **secant line** through the **point of tangency** and a **second point** on the curve.

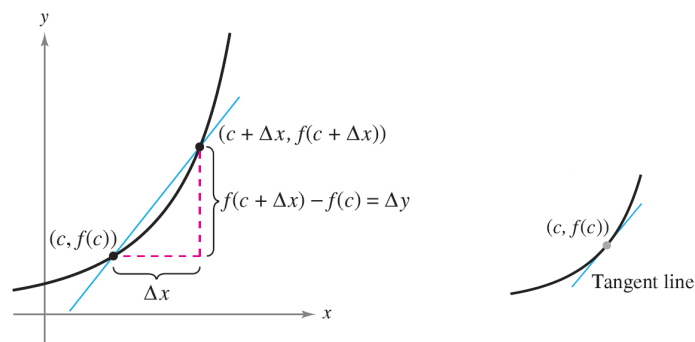


The Tangent Line Problem

- Obtain more and **more accurate approximations** of the slope of the tangent line by **choosing points closer and closer to the point of tangency**.



The Tangent Line Problem



DEFINITION OF TANGENT LINE WITH SLOPE m

If f is defined on an open interval containing c , and if the limit

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$

→ ura. iuvos tangent line

exists, then the line passing through $(c, f(c))$ with slope m is the **tangent line** to the graph of f at the point $(c, f(c))$. **the slope of the graph of f at $x = c$**

EXAMPLE 1 The Slope of the Graph of a Linear Function

Find the slope of the graph of

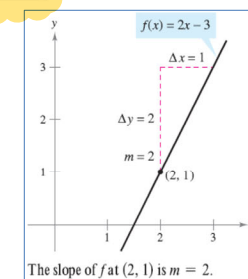
$$f(x) = 2x - 3$$

at the point $(2, 1)$.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$

Solution To find the slope of the graph of f when $c = 2$, you can apply the definition of the slope of a tangent line, as shown.

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{f(2 + \Delta x) - f(2)}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{[2(2 + \Delta x) - 3] - [2(2) - 3]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{4 + 2\Delta x - 3 - 4 + 3}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2\Delta x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} 2 \\ &= 2 \end{aligned}$$



The slope of f at $(c, f(c)) = (2, 1)$ is **$m = 2$**

EXAMPLE 2 Tangent Lines to the Graph of a Nonlinear Function

Find the slopes of the tangent lines to the graph of

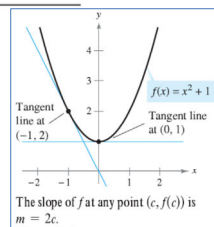
$$f(x) = x^2 + 1$$

at the points $(0, 1)$ and $(-1, 2)$, as shown in Figure

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$

Solution Let $(c, f(c))$ represent an arbitrary point on the graph of f . Then the slope of the tangent line at $(c, f(c))$ is given by

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{[(c + \Delta x)^2 + 1] - (c^2 + 1)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{c^2 + 2c(\Delta x) + (\Delta x)^2 + 1 - c^2 - 1}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2c(\Delta x) + (\Delta x)^2}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} (2c + \Delta x) \\ &= 2c. \end{aligned}$$



So, the slope at *any* point $(c, f(c))$ on the graph of f is $m = 2c$. At the point $(0, 1)$, the slope is $m = 2(0) = 0$, and at $(-1, 2)$, the slope is $m = 2(-1) = -2$.

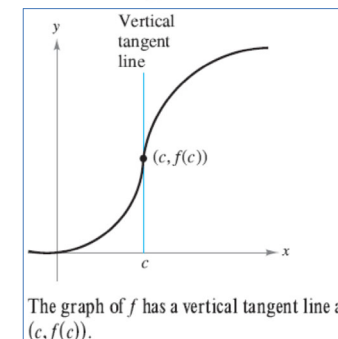
The Tangent Line Problem

- For **vertical tangent lines**:

– If f is continuous at c and

$$\lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = \infty \quad \text{or} \quad \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = -\infty$$

the vertical line $x = c$ passing through $(c, f(c))$ is a **vertical tangent line** to the graph of f .



The Derivative of a Function

DEFINITION OF THE DERIVATIVE OF A FUNCTION

The **derivative** of f at x is given by

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

provided the limit exists. For all x for which this limit exists, f' is a function of x .

gives the slope of the tangent line to the graph of f at the point $(x, f(x))$

The process of finding the derivative of a function is called **differentiation**. A function is **differentiable** at x if its derivative exists at x and is **differentiable on an open interval (a, b)** if it is differentiable at every point in the interval.

The Derivative of a Function

Derivative
of $f(x)$

" f prime of x ,"

$$f'(x), \frac{dy}{dx}, y', \frac{d}{dx}[f(x)], D_x[y].$$

Notation for derivatives

"the derivative of y with respect to x " or simply " dy, dx ."

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

$$\begin{aligned} &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ &= f'(x). \end{aligned}$$

EXAMPLE 3 Finding the Derivative by the Limit Process

Find the derivative of $f(x) = x^3 + 2x$.

Solution

EXAMPLE 3 Finding the Derivative by the Limit Process
Find the derivative of $f(x) = x^3 + 2x$.

Solution

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad \text{Definition of derivative}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^3 + 2(x + \Delta x) - (x^3 + 2x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3 + 2x + 2\Delta x - x^3 - 2x}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3 + 2\Delta x}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x[3x^2 + 3x(\Delta x) + (\Delta x)^2 + 2]}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} [3x^2 + 3x(\Delta x) + (\Delta x)^2 + 2]$$

$$= 3x^2 + 2$$

STUDY TIP When using the definition to find a derivative of a function, the key is to rewrite the difference quotient so that Δx does not occur as a factor of the denominator.

EXAMPLE 4 Using the Derivative to Find the Slope at a Point

Find $f'(x)$ for $f(x) = \sqrt{x}$. Then find the slopes of the graph of f at the points $(1, 1)$ and $(4, 2)$. Discuss the behavior of f at $(0, 0)$.

Solution

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \left(\frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x} \right) \left(\frac{\sqrt{x + \Delta x} + \sqrt{x}}{\sqrt{x + \Delta x} + \sqrt{x}} \right)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x) - x}{\Delta x (\sqrt{x + \Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x (\sqrt{x + \Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{1}{\sqrt{x + \Delta x} + \sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}, \quad x > 0$$

$$(1, 1) = \frac{1}{2\sqrt{1}} = \frac{1}{2} \quad \#$$

$$(4, 2) = \frac{1}{2\sqrt{4}} = \frac{1}{2(2)} = \frac{1}{4} \quad \#$$

$$(0, 0) = \text{undefined} \quad \#$$

Differentiability and Continuity



$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

Alternative form of derivative

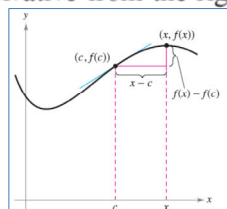
$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$

requires that the **one-sided limits**

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \quad \text{and} \quad \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$$

$\rightarrow$ $m \rightarrow$, $\leftarrow$ ค่าเท่ากันไหม
เท่ากัน = หาได้
ไม่เท่ากัน = หาไม่ได้

exist and are equal. These one-sided limits are called the **derivatives from the left** and **from the right**, respectively. It follows that f is **differentiable on the closed interval $[a, b]$** if it is differentiable on (a, b) and if the derivative from the right at a and the derivative from the left at b both exist.



As x approaches c , the secant line approaches the tangent line.

If a function is **not continuous at $x = c$** , it is **also not differentiable at $x = c$** . For instance, the greatest integer function

$$f(x) = \llbracket x \rrbracket$$

is not continuous at $x = 0$, and so it is not differentiable at $x = 0$

You can verify this by observing that

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{\llbracket x \rrbracket - 0}{x} = \infty$$

มือซ้าย: 0 เป็น ∞

and

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\llbracket x \rrbracket - 0}{x} = 0.$$

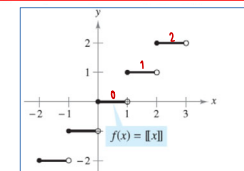
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Derivative from the left

$\downarrow$
หา 2 ข้าง ว่าเท่ากันไหม

Derivative from the right

Although it is true that **differentiability implies continuity** (as shown in Theorem 2.1 on the next page), **the converse is not true**. That is, **it is possible for a function to be continuous at $x = c$ and not differentiable at $x = c$** .



The greatest integer function is not differentiable at $x = 0$, because it is not continuous at $x = 0$.

EXAMPLE 6 A Graph with a Sharp Turn

NOT
differentiable

1/2

The function

$$f(x) = |x - 2|$$

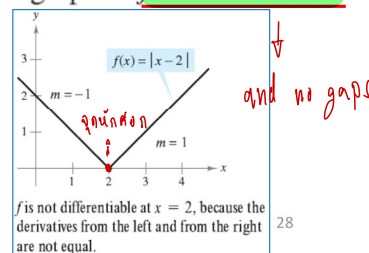
is continuous at $x = 2$. However, the one-sided limits

$$\lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2} = \lim_{x \rightarrow 2^-} \frac{|x - 2| - 0}{x - 2} = -1$$

and

$$\lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2} = \lim_{x \rightarrow 2^+} \frac{|x - 2| - 0}{x - 2} = 1$$

are not equal. So, f is not differentiable at $x = 2$ and the graph of f does not have a tangent line at the point $(2, 0)$.



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28

EXAMPLE 7 A Graph with a Vertical Tangent Line

NOT
differentiable

2/2

The function

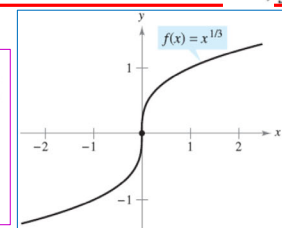
$$f(x) = x^{1/3}$$

is continuous at $x = 0$. However, because the limit

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} &= \lim_{x \rightarrow 0} \frac{x^{1/3} - 0}{x} \\ &= \lim_{x \rightarrow 0} \frac{1}{x^{2/3}} \\ &= \infty \end{aligned}$$

is infinite, you can conclude that the tangent line is vertical at $x = 0$. So, f is not differentiable at $x = 0$.

a function is not differentiable at a point at which its graph has a sharp turn or a vertical tangent line.



f is not differentiable at $x = 0$, because f has a vertical tangent line at $x = 0$.

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29

THEOREM 2.1 DIFFERENTIABILITY IMPLIES CONTINUITY

If f is differentiable at $x = c$, then f is continuous at $x = c$.

The following statements summarize the relationship between continuity and differentiability.

1. If a function is differentiable at $x = c$, then it is continuous at $x = c$. So, differentiability implies continuity.
2. It is possible for a function to be continuous at $x = c$ and not be differentiable at $x = c$. So, continuity does not imply differentiability.

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30

Basic Differentiation Rules and Rates of Change

Objectives:

Find

- the **derivative** of a function using the **Constant Rule**.
- the derivative of a function using the **Power Rule**.
- the derivative of a function using the **Constant Multiple Rule**.
- the derivative of a function using the **Sum and Difference Rules**.
- the **derivatives** of the **sine function** and of the **cosine function**.

Use

- derivatives to find **rates of change**.

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31

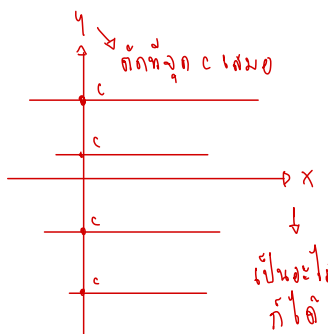
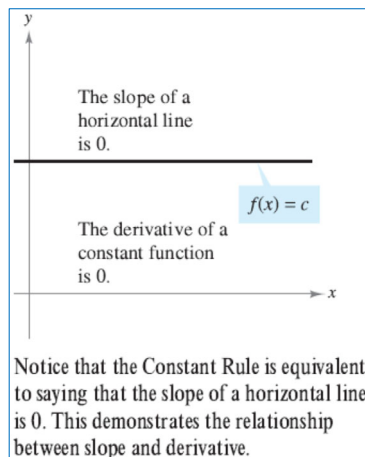
The Constant Rule

กฎค่าคงที่

THEOREM 2.2 THE CONSTANT RULE

The derivative of a constant function is 0. That is, if c is a real number, then

$$\frac{d}{dx}[c] = 0.$$



32

The Constant Rule

EXAMPLE 1 Using the Constant Rule

Function	Derivative
a. $y = 7$	$dy/dx = 0$
b. $f(x) = 0$	$f'(x) = 0$
c. $s(t) = -3$	$s'(t) = 0$
d. $y = k\pi^2$, k is constant	$y' = 0$

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33

The Power Rule

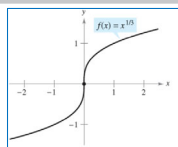
THEOREM 2.3 THE POWER RULE

If n is a rational number, then the function $f(x) = x^n$ is differentiable and

$$\frac{d}{dx}[x^n] = nx^{n-1}.$$

diff $\rightarrow$ คบค่าแล้วลบมาข้างหน้า, ซี่กค่าแล้ว -1

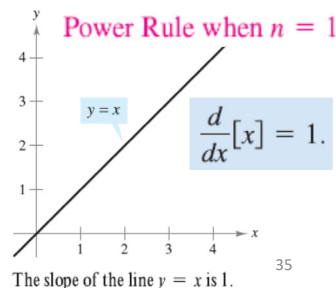
For f to be differentiable at $x = 0$, n must be a number such that x^{n-1} is defined on an interval containing 0.



NOTE

you know that the function $f(x) = x^{1/3}$ is defined at $x = 0$, but is not differentiable at $x = 0$. This is because $x^{-2/3}$ is not defined on an interval containing 0.

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35

The Power Rule

EXAMPLE 2 Using the Power Rule

Function	Derivative
a. $f(x) = x^{3-1} 3x^2$	$f'(x) = 3x^2$
b. $g(x) = \sqrt[3]{x} = x^{1/3}$	$g'(x) = \frac{d}{dx}[x^{1/3}] = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$
c. $y = \frac{1}{x^2} = x^{-2}$	$\frac{dy}{dx} = \frac{d}{dx}[x^{-2}] = (-2)x^{-3} = -\frac{2}{x^3}$

Given:
 $y = \frac{1}{x^2}$

Rewrite:
 $y = x^{-2}$

Differentiate:
 $\frac{dy}{dx} = (-2)x^{-3}$

Simplify:
 $\frac{dy}{dx} = -\frac{2}{x^3}$

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37

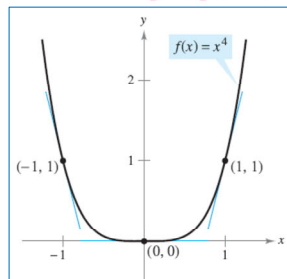
EXAMPLE 3 Finding the Slope of a Graph

Find the slope of the graph of $f(x) = x^4$ when $f'(x) = 4x^3$

- a. $x = -1$ b. $x = 0$ c. $x = 1$.

Solution The slope of a graph at a point is the value of the derivative at that point. The derivative of f is $f'(x) = 4x^3$.

- a. When $x = -1$, the slope is $f'(-1) = 4(-1)^3 = -4$. Slope is negative.
 b. When $x = 0$, the slope is $f'(0) = 4(0)^3 = 0$. Slope is zero.
 c. When $x = 1$, the slope is $f'(1) = 4(1)^3 = 4$. Slope is positive.



Note that the slope of the graph is negative at the point $(-1, 1)$, the slope is zero at the point $(0, 0)$, and the slope is positive at the point $(1, 1)$.

* ก. จัน → อ. ตา ก. เปลี่ยนแปลง

EXAMPLE 4 Finding an Equation of a Tangent Line

Find an equation of the tangent line to the graph of $f(x) = x^2$ when $x = -2$.

Solution

$$y = (-2)^2 = 4$$

$$x = -2, y = 4$$

∴ $(-2, 4)$ is point of graph

$$2(-2) = -4$$

↓
Slope at $(-2, 4)$

ใช้สูตรเส้นสัมผัส

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -4[x - (-2)]$$

$$y = -4x - 4$$

การใช้จุด / slope ที่จุดนั้น

Point-slope form



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m(x_2 - x_1) = y_2 - y_1$$

BREAK

The Constant Multiple Rule

THEOREM 2.4 THE CONSTANT MULTIPLE RULE

If f is a differentiable function and c is a real number, then cf is also differentiable and $\frac{d}{dx}[cf(x)] = cf'(x)$.

$$\frac{d}{dx}[cf(x)] = c \frac{d}{dx}[f(x)] = cf'(x)$$

$$\frac{d}{dx}\left[\frac{f(x)}{c}\right] = \frac{d}{dx}\left[\left(\frac{1}{c}\right)f(x)\right]$$

$$= \left(\frac{1}{c}\right) \frac{d}{dx}[f(x)] = \left(\frac{1}{c}\right)f'(x)$$

EXAMPLE 5 Using the Constant Multiple Rule

Function	Derivative
a. $y = \frac{2}{x} = 2x^{-1}$	$\frac{dy}{dx} = 2(-1)(x^{-2}) = -2(x^{-2}) = -\frac{2}{x^2}$
b. $f(t) = \frac{4t^2}{5}$	$f'(t) = \frac{4}{5}t_2 = \frac{4}{5}(2t) = \frac{8}{5}t$
c. $y = 2\sqrt{x}$	$\frac{dy}{dx} = 2x^{\frac{1}{2}} = 2(\frac{1}{2}x^{-\frac{1}{2}}) = x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$
d. $y = \frac{1}{2\sqrt[3]{x^2}}$	$\frac{dy}{dx} = \frac{1}{2}(x^{-2/3}) = \frac{1}{2}(\frac{-2}{3}x^{-\frac{5}{3}}) = -\frac{1}{3x^{5/3}}$
e. $y = -\frac{3x}{2}$	$y' = -\frac{3}{2}$

The Constant Multiple Rule and the Power Rule can be combined into one rule. The combination rule is

$$\frac{d}{dx}[cx^n] = cnx^{n-1}.$$

EXAMPLE 6 Using Parentheses When Differentiating

Original Function	Rewrite	Differentiate	Simplify
a. $y = \frac{5}{2x^3}$	$y = \frac{5}{2}(x^{-3})$	$y' = \frac{5}{2}(-3x^{-4})$	$y' = -\frac{15}{2x^4}$
b. $y = \frac{5}{(2x)^3}$	$y = \frac{5x^{-3}}{8}$	$y' = \frac{5}{8}(-3x^{-4})$	$y' = \frac{5}{8}(\frac{-3}{x^4}) = -\frac{15}{8x^4}$
c. $y = \frac{7}{3x^{-2}}$	$y = \frac{7x^2}{3}$	$y' = \frac{14x}{3}$	
d. $y = \frac{7}{(3x)^{-2}}$	$y = \frac{7}{-9x^{-2}} = -\frac{7}{9}x^2$	$y' = -\frac{7}{9}(2x) = -\frac{14}{9}x$	

The Sum and Difference Rules

THEOREM 2.5 THE SUM AND DIFFERENCE RULES

The sum (or difference) of two differentiable functions f and g is itself differentiable. Moreover, the derivative of $f + g$ (or $f - g$) is the sum (or difference) of the derivatives of f and g .

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x) \quad \text{Sum Rule}$$

$$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x) \quad \text{Difference Rule}$$

EXAMPLE 7 Using the Sum and Difference Rules

Function	Derivative
a. $f(x) = x^3 - 4x + 5$	$3x^2 - 4 + 0 = 3x^2 - 4$
b. $g(x) = -\frac{x^4}{2} + 3x^3 - 2x$	$-\frac{4x^3}{2} + 9x^2 - 2 = -2x^3 + 9x^2 - 2$

Derivatives of the Sine and Cosine Functions

THEOREM 2.6 DERIVATIVES OF SINE AND COSINE FUNCTIONS

$$\frac{d}{dx}[\sin x] = \cos x$$

$$\frac{d}{dx}[\cos x] = -\sin x$$

EXAMPLE 8 Derivatives Involving Sines and Cosines

Function	Derivative
a. $y = 2 \sin x$	$y' = 2 \cos x$
b. $y = \frac{\sin x}{2} = \frac{1}{2} \sin x$	$y' = \frac{1}{2} \cos x$
c. $y = x + \cos x$	$y' = 1 - \sin x$

Rates of Change

- The **derivative** can also be used to determine the **rate of change** of one variable with respect to another.
- E.g.,
 - Population growth rates
 - Production rates
 - Water flow rates
 - Velocity
 - Acceleration

Rates of Change

- A common use for rate of change is to describe the **motion** of an object moving **in a straight line**.
- The function s that gives the position (relative to the origin) of an object as a **function of time t** is called a **position function**.

$$\text{Rate} = \frac{\text{distance}}{\text{time}}$$

the **average velocity** is

$$\frac{\text{Change in distance}}{\text{Change in time}} = \frac{\Delta s}{\Delta t}$$

Average velocity

EXAMPLE 9 Finding Average Velocity of a Falling Object

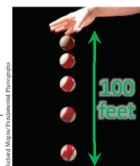
If a billiard ball is dropped from a height of 100 feet, its height s at time t is given by the position function

$$s = -16t^2 + 100$$

Position function

where s is measured in feet and t is measured in seconds. Find the average velocity over each of the following time intervals.

- a. $[1, 2]$ b. $[1, 1.5]$ c. $[1, 1.1]$



Solution

- a. For the interval $[1, 2]$, the object falls from a height of $s(1) = -16(1)^2 + 100 = 84$ feet to a height of $s(2) = -16(2)^2 + 100 = 36$ feet. The average velocity is

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$$\frac{\Delta s}{\Delta t} = \frac{36 - 84}{2 - 1} = -48 \text{ feet per second}$$

EXAMPLE 9 Finding Average Velocity of a Falling Object

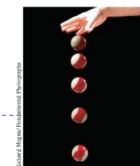
If a billiard ball is dropped from a height of 100 feet, its height s at time t is given by the position function

$$s = -16t^2 + 100$$

Position function

where s is measured in feet and t is measured in seconds. Find the average velocity over each of the following time intervals.

- a. $[1, 2]$ b. $[1, 1.5]$ c. $[1, 1.1]$



Solution

b.

$$\frac{\Delta s}{\Delta t} = \frac{64 - 84}{1.5 - 1} = \frac{-20}{0.5} = -40 \text{ ft/s}$$



EXAMPLE 9 Finding Average Velocity of a Falling Object

If a billiard ball is dropped from a height of 100 feet, its height s at time t is given by the position function

$$s = -16t^2 + 100$$

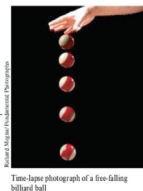
Position function

where s is measured in feet and t is measured in seconds. Find the average velocity over each of the following time intervals.

- a. $[1, 2]$ b. $[1, 1.5]$ c. $[1, 1.1]$

Solution

c.



In general, if $s = s(t)$ is the position function for an object moving along a straight line, the **velocity** of the object at time t is

ก. หา derivative func.

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$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t} = s'(t).$$

Velocity function

In other words, the velocity function is the derivative of the position function. Velocity can be negative, zero, or positive. The **speed** of an object is the absolute value of its velocity. Speed cannot be negative.

The position of a free-falling object (neglecting air resistance) under the influence of gravity can be represented by the equation

$$s(t) = \frac{1}{2}gt^2 + v_0t + s_0$$

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ค. เปรียบจากความเร็วโน้มถ่วง (9.8 m/s²) Position function

where s_0 is the initial height of the object, v_0 is the initial velocity of the object, and g is the acceleration due to gravity. On Earth, the value of g is approximately -32 feet per second per second or -9.8 meters per second per second.

EXAMPLE 10 Using the Derivative to Find Velocity

At time $t = 0$, a diver jumps from a platform diving board that is 32 feet above the water. The position of the diver is given by

$$s(t) = -16t^2 + 16t + 32$$

Position function

where s is measured in feet and t is measured in seconds.

- a. When does the diver hit the water?
b. What is the diver's velocity at impact?

Solution

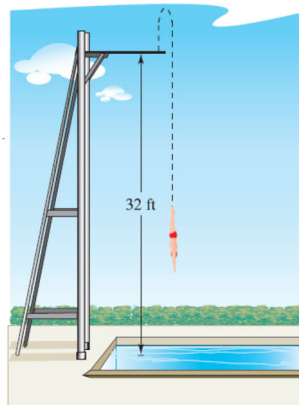


$$\begin{aligned} -16t^2 + 16t + 32 &= 0 \\ -16(t+1)(t-2) &= 0 \end{aligned}$$

$t = -1$ or 2 positive

Ans $t = 2$

$$\begin{aligned} \text{b. } s'(t) &= -32t + 16; \quad t = 2; \\ s'(2) &= -32(2) + 16 = -48 \text{ f/s} \\ \text{Ans } v &= -48 \text{ f/s} \end{aligned}$$



Velocity is positive when an object is rising, and is negative when an object is falling. Notice that the diver moves upward for the first half-second because the velocity is positive for $0 < t < \frac{1}{2}$. When the velocity is 0, the diver has reached the maximum height of the dive.

Product and Quotient Rules

Objectives

- Find the derivative of a function using the Product Rule.
- Find the derivative of a function using the Quotient Rule.
- Find the derivative of a trigonometric function.

The Product Rule

THEOREM 2.7 THE PRODUCT RULE

The product of two differentiable functions f and g is itself differentiable. Moreover, the derivative of fg is the first function times the derivative of the second, plus the second function times the derivative of the first.

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x) \quad \text{ฟังก์ชันแรก}(\text{อนุพันธ์ของฟังก์ชันที่สอง}) + \text{ฟังก์ชันที่สอง}(\text{อนุพันธ์ของฟังก์ชันแรก})$$

The Product Rule can be extended to cover products involving more than two factors. For example, if f , g , and h are differentiable functions of x , then

$$\frac{d}{dx}[f(x)g(x)h(x)] = f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x).$$

EXAMPLE 2 Using the Product Rule

Find the derivative of $y = 3x^2 \sin x$.

Solution



EXAMPLE 1 Using the Product Rule

Find the derivative of $h(x) = (3x - 2x^2)(5 + 4x)$.

Solution

$$\begin{aligned} h'(x) &= \overbrace{(3x - 2x^2)}^{\text{First}} \overbrace{\frac{d}{dx}[5 + 4x]}^{\text{Derivative of second}} + \overbrace{(5 + 4x)}^{\text{Second}} \overbrace{\frac{d}{dx}[3x - 2x^2]}^{\text{Derivative of first}} && \text{Apply Product Rule.} \\ &= (3x - 2x^2)(4) + (5 + 4x)(3 - 4x) \\ &= (12x - 8x^2) + (15 - 8x - 16x^2) \\ &= -24x^2 + 4x + 15 \end{aligned}$$

In Example 1, you have the option of finding the derivative with or without the Product Rule. To find the derivative without the Product Rule, you can write

$$\begin{aligned} D_x[(3x - 2x^2)(5 + 4x)] &= D_x[-8x^3 + 2x^2 + 15x] \\ &= -24x^2 + 4x + 15. \end{aligned}$$

EXAMPLE 3 Using the Product Rule

Find the derivative of $y = 2x \cos x - 2 \sin x$.

Solution



EXAMPLE 3 Using the Product Rule

Find the derivative of $y = 2x \cos x - 2 \sin x$.

Solution

$$\begin{aligned} \frac{dy}{dx} &= (2x) \left(\frac{d}{dx} [\cos x] \right) + (\cos x) \left(\frac{d}{dx} [2x] \right) - 2 \frac{d}{dx} [\sin x] \\ &= (2x)(-\sin x) + (\cos x)(2) - 2(\cos x) \\ &= -2x \sin x \\ &= -2 \sin x \end{aligned}$$

NOTE In Example 3, notice that you use the Product Rule when both factors of the product are variable, and you use the Constant Multiple Rule when one of the factors is a constant.

The Quotient Rule

THEOREM 2.8 THE QUOTIENT RULE

The quotient f/g of two differentiable functions f and g is itself differentiable at all values of x for which $g(x) \neq 0$. Moreover, the derivative of f/g is given by the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator.

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}, \quad g(x) \neq 0$$

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2}$$

EXAMPLE 4 Using the Quotient Rule

Find the derivative of $y = \frac{5x - 2}{x^2 + 1}$.

Solution



EXAMPLE 4 Using the Quotient Rule

Find the derivative of $y = \frac{5x - 2}{x^2 + 1}$.

Solution

$$\begin{aligned} \frac{d}{dx} \left[\frac{5x - 2}{x^2 + 1} \right] &= \frac{(x^2 + 1) \frac{d}{dx} [5x - 2] - (5x - 2) \frac{d}{dx} [x^2 + 1]}{(x^2 + 1)^2} \\ &= \frac{(x^2 + 1)(5) - (5x - 2)(2x)}{(x^2 + 1)^2} \\ &= \frac{(5x^2 + 5) - (10x^2 - 4x)}{(x^2 + 1)^2} \\ &= \frac{-5x^2 + 4x + 5}{(x^2 + 1)^2} \end{aligned}$$

Math for IT (Somkiat Wangsiripitak) 71

EXAMPLE 6 Using the Constant Multiple Rule

Original Function	Rewrite	Differentiate	Simplify
a. $y = \frac{x^2 + 3x}{6}$	$\frac{1}{6}(x^2 + 3x)$	$\frac{1}{6}(2x + 3)$	$\frac{2x + 3}{6}$
b. $y = \frac{5x^4}{8}$		$\frac{10 \cdot 5x^3}{8 \cdot 4} = \frac{5 \cdot 20x^3}{4 \cdot 2}$	$= \frac{5}{2}x^3$
c. $y = \frac{-3(3x - 2x^2)}{7x}$			
d. $y = \frac{9}{5x^2}$			

NOTE To see the benefit of using the Constant Multiple Rule for some quotients, try using the Quotient Rule to differentiate the functions in Example 6—you should obtain the same results, but with more work.

Derivatives of Trigonometric Functions

THEOREM 2.9 DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

$$\frac{d}{dx} [\tan x] = \sec^2 x$$

$$\frac{d}{dx} [\cot x] = -\csc^2 x$$

$$\frac{d}{dx} [\sec x] = \sec x \tan x$$

$$\frac{d}{dx} [\csc x] = -\csc x \cot x$$

EXAMPLE 8 Differentiating Trigonometric Functions

Function	Derivative
a. $y = x - \tan x$	$\frac{dy}{dx} = 1 - \sec^2 x$
b. $y = x \sec x$	$y' = x(\sec x \tan x) + (\sec x)(1)$ $= (\sec x)(1 + x \tan x)$

NOTE Because of trigonometric identities, the derivative of a trigonometric function can take many forms. This presents a challenge when you are trying to match your answers to those given in the back of the text.

$$\sin \theta = \frac{1}{\csc \theta}$$

↓
csc θ

EXAMPLE 9 Different Forms of a Derivative

Differentiate both forms of $y = \frac{1 - \cos x}{\sin x} = \csc x - \cot x$.

Solution

First form: $y = \frac{1 - \cos x}{\sin x}$

$$y' = \frac{(\sin x)(\sin x) - (1 - \cos x)(\cos x)}{\sin^2 x}$$

$$= \frac{\sin^2 x + \cos^2 x - \cos x}{\sin^2 x}$$

$$= \frac{1 - \cos x}{\sin^2 x}$$

Second form: $y = \csc x - \cot x$

$$y' = -\csc x \cot x + \csc^2 x$$

To show that the two derivatives are equal, you can write

$$\frac{1 - \cos x}{\sin^2 x} = \frac{1}{\sin^2 x} - \left(\frac{1}{\sin x} \right) \left(\frac{\cos x}{\sin x} \right)$$

$$= \csc^2 x - \csc x \cot x.$$

Homework

Find the derivative of the function below.

$$g(x) = (x^3 + 7x)(x + 3)$$

$$f(x) = \frac{6x-5}{x^2+1}$$

$$y = \frac{\sin x}{x^4}$$

- Write down a step-by-step solution in a piece of A4 paper (or on an electronic paper).
- **DO NOT TYPE.**
- Submit your solution in MS Team.
- Do not forget to write your name and student ID.